

AI QB UNIT 6 (4 marks)

1. Define Natural Language Processing (NLP)

1. Natural Language Processing (NLP) is a branch of Artificial Intelligence that helps computers understand and process human language. It combines computer science, linguistics, and machine learning for better communication between humans and machines.
2. NLP enables computers to understand text and speech data effectively. It also helps systems generate meaningful responses to users.
3. The main objective of NLP is to make human-computer interaction easier and smarter. It allows machines to process language in a natural way.
4. NLP is widely used in chatbots and virtual assistants. It is also used in speech recognition and machine translation systems.
5. NLP helps in sentiment analysis and text summarization applications. It is useful for analyzing large amounts of language data quickly.
6. NLP plays an important role in modern AI applications. It improves communication between humans and computers in many fields.

2. What is a language model in NLP?

1. A language model is a system that predicts the probability of occurrence of words in a sentence. It helps computers understand language patterns and generate meaningful text.
2. Language models are used to predict the next word in a sentence. They are also useful in speech recognition and text generation tasks.
3. Statistical language models use probability calculations based on word sequences. An example of this type is the N-gram model.
4. Neural language models use deep learning methods for better understanding of text. These models provide better accuracy and context understanding.
5. Examples of neural language models include RNN, LSTM, and Transformers. These models are commonly used in modern NLP systems.
6. Language models help computers generate meaningful predictions for incomplete sentences. For example, "She is drinking ____" may predict water, coffee, or juice.

3. Define grammar in NLP

1. Grammar in NLP refers to a set of rules that define how words combine to form valid sentences. It helps computers understand sentence structure correctly.
2. Grammar is useful for sentence formation and syntax checking. It also helps in proper language understanding by computers.
3. Regular grammar is the simplest type of grammar used in NLP. It is mainly used in lexical analysis and simple word patterns.
4. Context-Free Grammar (CFG) is one of the most widely used grammars in NLP. It uses rules such as $S \rightarrow NP + VP$ for sentence formation.
5. Context-sensitive grammar depends on the context of words in a sentence. It is used for handling more complex language structures.
6. Dependency grammar focuses on relationships between words in a sentence. It helps identify how words depend on each other for meaning.

4. What is parsing in NLP?

1. Parsing in NLP is the process of analyzing a sentence according to grammar rules. It helps determine the structure and meaning of the sentence.
2. Parsing helps identify important parts of a sentence such as subject, verb, and object. It also finds relationships between different phrases in a sentence.
3. Parsing is important for sentence understanding in NLP systems. It is also used in machine translation and question answering systems.
4. Top-down parsing starts from the root symbol and breaks the sentence into smaller parts. It follows grammar rules step by step to analyze the sentence.
5. Bottom-up parsing starts from input words and builds the sentence structure gradually. Dependency parsing focuses on relationships between words in the sentence.
6. Constituency parsing divides a sentence into meaningful phrases. For example, "[NP The boy] [VP plays football]" shows noun phrase and verb phrase structure.

5. Define robotics

1. Robotics is a branch of Artificial Intelligence and engineering that deals with the design and development of robots. It involves creating machines that can perform tasks automatically.
2. Robotics combines computer science, electronics, and mechanical engineering technologies. It helps robots perform tasks with speed, accuracy, and less human effort.
3. Robots are machines programmed to perform specific actions automatically. They can work in industries, hospitals, homes, and research fields.
4. Robotics is used in manufacturing, medical surgery, space research, and defense applications. It improves productivity and reduces human workload in dangerous tasks.
5. Robots can sense their environment using sensors and respond using programmed instructions. Some advanced robots can also use AI techniques for decision-making.
6. Robotics plays an important role in automation and modern technology systems. It helps industries perform repetitive and complex tasks efficiently.

6. What are robot hardware components?

1. Robot hardware components are the physical parts used to build and operate a robot. These components help the robot sense, process, and perform actions.
2. Sensors are important robot components that collect information from the environment. They help robots detect light, temperature, distance, and movement.
3. Actuators are components that help robots perform physical movements. Motors and hydraulic systems are common examples of actuators.
4. The controller is the brain of the robot that processes instructions and controls operations. It manages the actions of all robot components.
5. Power supply components provide energy required for robot operation. Batteries and electrical power systems are commonly used in robots.
6. Mechanical structures such as wheels, arms, and joints help robots move and perform tasks. These parts provide stability and physical support to the robot.

(5 – marks)

1. Explain language models in NLP with example

1. A language model in NLP is a system that predicts the probability of words occurring in a sentence. It helps computers understand language patterns and generate meaningful text.
2. Language models are used for predicting the next word in a sentence. They are also useful in speech recognition, text generation, and machine translation systems.
3. Statistical language models use probability calculations based on sequences of words. An example of this type is the N-gram model.
4. In a bigram model, the next word prediction depends on the previous word. For example, in the sentence "I am going to the market", the word "market" depends on "the".
5. Neural language models use deep learning techniques for better understanding of language. These models provide higher accuracy and better context understanding.
6. Examples of neural models include RNN, LSTM, and Transformers. These models are widely used in modern NLP systems like chatbots and translators.
7. Language models can predict missing words in sentences. For example, "She is drinking ____" may predict water, coffee, or juice.

2. Explain grammars used in NLP

1. Grammar in NLP refers to a set of rules that define how words combine to form valid sentences. It helps in sentence formation, syntax checking, and language understanding.
2. Regular grammar is the simplest type of grammar used in NLP. It is mainly used in lexical analysis and handling simple word patterns.
3. Context-Free Grammar (CFG) is the most widely used grammar in NLP systems. It uses rules like $S \rightarrow NP + VP$ for sentence formation.
4. In CFG, S represents a sentence, NP represents a noun phrase, and VP represents a verb phrase. For example, in "Ram eats mango", Ram is NP and eats mango is VP.
5. Context-sensitive grammar depends on the context of words in a sentence. It is used for analyzing more complex language structures.
6. Dependency grammar focuses on relationships between words in a sentence. It identifies how words depend on each other for meaning.
7. Grammars are important in NLP because they help computers understand sentence structure correctly. They improve language analysis and sentence interpretation in AI systems.

3. Explain parsing techniques in NLP

1. Parsing in NLP is the process of analyzing a sentence according to grammar rules. It helps determine the structure and meaning of a sentence.
2. Parsing identifies important parts of a sentence such as subject, verb, and object. It also helps find relationships between words and phrases.
3. Top-down parsing starts from the root symbol and breaks the sentence into smaller parts. It follows grammar rules step by step to analyze the sentence structure.
4. Bottom-up parsing starts from the input words and gradually builds the sentence structure. It constructs the parse tree from smaller units to larger units.
5. Dependency parsing focuses on dependency relationships between words in a sentence. For example, in “The boy plays football”, boy is the subject and football is the object.
6. Constituency parsing divides a sentence into meaningful phrases such as noun phrase and verb phrase. Example: “[NP The boy] [VP plays football]”.
7. Parsing is important in NLP applications like machine translation, speech processing, and question answering systems. It helps computers understand sentence meaning more accurately.

4. Describe robot hardware architecture

1. Robot hardware architecture refers to the physical structure and components of a robot. It includes all parts required for sensing, processing, movement, and control.
2. Sensors are important components that collect information from the surrounding environment. They help robots detect temperature, light, distance, and obstacles.
3. Controllers act as the brain of the robot and process all instructions. They control the working of sensors, actuators, and other robot parts.
4. Actuators help robots perform physical movements and actions. Motors, hydraulic systems, and pneumatic systems are common types of actuators.
5. The power supply provides energy required for robot operation and movement. Batteries and electrical power systems are widely used in robots.
6. Mechanical structures such as arms, wheels, joints, and frames provide support and movement to robots. These structures help robots perform different tasks efficiently.
7. Communication systems help robots exchange data with users or other machines. This architecture allows robots to work automatically and accurately in many applications.

5. Explain problems that robotics can solve

1. Robotics helps solve problems involving dangerous and risky tasks for humans. Robots can safely work in mines, nuclear plants, and disaster areas.
2. Robots can perform repetitive tasks with high speed and accuracy. This reduces human workload and improves industrial productivity.
3. Robotics is useful in healthcare for surgeries and patient assistance. Medical robots help doctors perform operations more precisely.
4. Robots can work continuously without getting tired or making many mistakes. This improves efficiency and saves time in industries and factories.
5. Robotics helps in space exploration and underwater research where humans cannot easily work. Robots can collect data and perform tasks in difficult environments.
6. Service robots help solve daily life problems such as cleaning, delivery, and security monitoring. They improve convenience and automation in homes and offices.
7. Robotics also helps in agriculture by automating farming activities like watering, harvesting, and crop monitoring. This increases productivity and reduces manual labor.

6. Explain AI applications in healthcare

1. Artificial Intelligence is widely used in healthcare to improve medical services and patient care. It helps doctors analyze medical data quickly and accurately.
2. AI systems assist in disease diagnosis by analyzing medical reports and images. They help detect diseases such as cancer and heart problems at an early stage.
3. AI is used in robotic surgery to improve precision during operations. Surgical robots help doctors perform complex surgeries more safely.
4. Virtual assistants and chatbots provide healthcare support to patients online. They answer medical questions and help schedule appointments.
5. AI helps hospitals manage patient records and healthcare data efficiently. It reduces paperwork and improves data management systems.
6. AI applications are used in drug discovery and medical research. They help scientists analyze large amounts of data faster than humans.
7. AI also supports remote healthcare and telemedicine services for patients in distant areas. This improves access to medical treatment and healthcare facilities.

7. Explain AI applications in retail

1. Artificial Intelligence is widely used in the retail industry to improve customer experience and business operations. It helps stores understand customer behavior and preferences more effectively.
2. AI systems provide personalized product recommendations based on customer purchase history and interests. This helps increase sales and customer satisfaction.
3. Chatbots and virtual assistants are used in online shopping platforms for customer support. They answer customer questions and provide quick assistance at any time.
4. AI helps retailers manage inventory and stock levels automatically. It reduces wastage and ensures products are available when needed.
5. AI-powered systems analyze customer data and market trends for better decision-making. This helps retailers plan pricing, offers, and marketing strategies.
6. Facial recognition and smart surveillance systems improve security in retail stores. These systems help detect theft and monitor store activities efficiently.
7. AI is also used in automated billing and cashier-less shopping systems. This reduces waiting time and improves the shopping experience for customers.

8. Explain AI applications in banking

1. Artificial Intelligence is used in banking to improve customer service and financial operations. It helps banks process large amounts of data quickly and accurately.
2. AI-powered chatbots and virtual assistants provide 24/7 customer support in banks. They help customers check balances, transfer money, and solve queries easily.
3. AI helps banks detect fraudulent transactions and suspicious activities. It improves security by identifying unusual patterns in banking operations.
4. AI systems are used for credit scoring and loan approval processes. They analyze customer financial history and reduce the time required for decision-making.
5. AI helps banks manage risk and predict financial trends using data analysis. This supports better investment and financial planning decisions.
6. Automated systems powered by AI reduce paperwork and manual banking tasks. They improve efficiency and reduce operational costs for banks.
7. AI is also used in personalized banking services and financial recommendations. It helps banks provide better services based on customer needs and behavior.

(10 -marks)

1. Explain Natural Language Processing (NLP) in detail with language models, grammars, and parsing

1. Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language. It combines computer science, linguistics, and machine learning for effective communication between humans and machines.
2. NLP is used in applications such as chatbots, machine translation, speech recognition, and sentiment analysis. It helps computers process text and speech data efficiently.
3. A language model is a system that predicts the probability of occurrence of words in a sentence. It helps computers understand language patterns and generate meaningful text.
4. Language models are mainly divided into statistical and neural language models. Examples of neural models include RNN, LSTM, and Transformers.
5. Grammar in NLP refers to a set of rules that define how words combine to form valid sentences. It helps in sentence formation, syntax checking, and language understanding.
6. Different types of grammars include regular grammar, context-free grammar, context-sensitive grammar, and dependency grammar. Context-Free Grammar uses rules like $S \rightarrow NP + VP$ for sentence formation.
7. Parsing is the process of analyzing a sentence according to grammar rules to determine its structure and meaning. It helps identify subjects, verbs, objects, and phrase relationships.
8. Types of parsing include top-down parsing, bottom-up parsing, dependency parsing, and constituency parsing. These methods help computers understand sentence structure correctly.
9. The NLP pipeline includes steps such as tokenization, stop-word removal, stemming, grammar analysis, parsing, and semantic analysis. These processes help computers understand the actual meaning of language.
10. NLP plays an important role in modern AI applications and improves human-computer interaction. It enables systems like translators, voice assistants, and intelligent chatbots to work effectively.

2. Explain robotics in detail including robot structure, hardware, and applications

1. Robotics is a branch of Artificial Intelligence and engineering that deals with the design, development, and operation of robots. It helps machines perform tasks automatically with accuracy and speed.
2. A robot is a programmable machine that can sense its environment and perform actions. Robots are widely used in industries, healthcare, agriculture, and research fields.
3. The structure of a robot includes mechanical parts such as arms, wheels, joints, and frames. These parts provide movement, support, and flexibility to the robot.
4. Sensors are important hardware components that help robots collect information from the environment. They can detect temperature, light, pressure, distance, and obstacles.
5. Controllers act as the brain of the robot and process all instructions. They control the working of sensors, actuators, and other hardware components.
6. Actuators help robots perform physical movements and actions efficiently. Motors, hydraulic systems, and pneumatic systems are common examples of actuators.
7. Power supply systems provide the energy required for robot operation. Batteries and electrical systems are commonly used to run robots.
8. Robotics is used in manufacturing industries for welding, painting, packaging, and assembly work. Robots increase productivity and reduce human effort in repetitive tasks.
9. Medical robots help doctors perform surgeries with high precision and safety. Robots are also used in space exploration, underwater research, and defense applications.
10. Robotics plays an important role in automation and modern technology systems. It improves efficiency, accuracy, and safety in many real-world applications.

3. Discuss problems solved by robotics with real-world examples

1. Robotics helps solve dangerous and risky problems where human safety is important. Robots are used in mines, nuclear plants, and disaster areas to reduce human risk.
2. Robots solve repetitive industrial problems by performing tasks quickly and accurately. Manufacturing industries use robots for assembly, welding, and packaging operations.
3. Robotics helps improve healthcare services through robotic surgeries and patient care systems. Surgical robots assist doctors in performing complex operations with greater precision.
4. Robots solve problems in space exploration where humans cannot easily survive or travel. Space agencies use robots and rovers to collect data from planets and space environments.
5. Robotics is useful in agriculture for automating farming activities like watering, harvesting, and crop monitoring. Agricultural robots help increase productivity and reduce manual labor.
6. Robots solve cleaning and maintenance problems in homes, offices, and industries. Vacuum cleaning robots and automated maintenance systems save time and human effort.
7. Robotics helps improve security and surveillance in public and private places. Security robots monitor activities and detect suspicious behavior using sensors and cameras.
8. Delivery and service robots solve transportation and customer service problems efficiently. Restaurants, hospitals, and warehouses use robots for delivery tasks and assistance.
9. Underwater robots help solve research and inspection problems in deep oceans where humans cannot work safely. These robots are used for underwater exploration and pipeline inspection.
10. Robotics improves efficiency, accuracy, and safety in many real-world situations. It helps humans solve complex problems faster and with fewer errors.

4. Explain AI applications in healthcare with case studies

1. Artificial Intelligence is widely used in healthcare to improve diagnosis, treatment, and patient care. It helps doctors analyze medical data quickly and accurately.
2. AI systems are used for disease diagnosis through medical image analysis and patient records. They help detect diseases such as cancer, diabetes, and heart problems at early stages.
3. A real-world example is the use of AI in cancer detection systems. AI-based tools analyze X-rays and MRI scans to identify tumors more accurately and quickly.
4. AI-powered robotic surgery systems help doctors perform complex surgeries with high precision. These systems reduce surgical errors and improve patient recovery speed.
5. Chatbots and virtual assistants provide healthcare support and answer patient queries online. They also help patients schedule appointments and receive medical guidance easily.
6. AI is used in drug discovery and medical research for faster analysis of large datasets. It helps scientists develop medicines and treatments more efficiently.
7. Hospitals use AI systems to manage patient records and healthcare operations automatically. This reduces paperwork and improves hospital management efficiency.
8. Telemedicine applications powered by AI help doctors provide remote treatment to patients. This is especially useful for people living in rural or distant areas.
9. During the COVID-19 pandemic, AI was used to track infection spread and analyze patient data. It helped healthcare systems make faster and better decisions.
10. AI applications in healthcare improve accuracy, efficiency, and patient safety in medical services. They support doctors and healthcare workers in providing better treatment.

5. Explain AI applications in retail and banking with examples

1. Artificial Intelligence is widely used in retail and banking industries to improve customer service and business operations. It helps organizations process large amounts of data efficiently.
2. In retail, AI systems provide personalized product recommendations based on customer interests and purchase history. Online shopping platforms use this feature to increase sales and customer satisfaction.
3. Chatbots and virtual assistants are commonly used in retail websites for customer support services. They answer customer questions and provide assistance at any time.
4. AI helps retailers manage inventory and stock levels automatically. This reduces product shortages and avoids unnecessary wastage.
5. In banking, AI-powered chatbots help customers perform tasks such as checking balances and transferring money. These systems provide fast and continuous customer support.
6. AI systems are used in banks to detect fraud and suspicious financial activities. They identify unusual transaction patterns and improve banking security.
7. Banks use AI for credit scoring and loan approval processes. AI analyzes customer financial records and helps make faster decisions.
8. A real-world example in retail is cashier-less smart stores that use AI for automated billing systems. Customers can shop without waiting in long billing lines.
9. A real-world example in banking is fraud detection systems used by banks and online payment platforms. These systems help prevent financial fraud and cybercrime.
10. AI applications in retail and banking improve efficiency, accuracy, security, and customer experience. They help businesses provide faster and smarter services.

6. Discuss overall impact of AI applications in real-world domains

1. Artificial Intelligence has created major changes in healthcare, education, banking, retail, and transportation sectors. It helps automate tasks and improve decision-making processes.
2. AI applications improve efficiency and accuracy in many industries by reducing human errors. They help organizations complete tasks faster and more effectively.
3. In healthcare, AI supports disease diagnosis, robotic surgery, and remote patient monitoring systems. This improves patient care and medical treatment quality.
4. In banking and retail, AI improves customer service through chatbots, recommendation systems, and fraud detection tools. These applications increase customer satisfaction and security.
5. AI is also used in transportation for self-driving vehicles and traffic management systems. It helps improve road safety and reduce travel time.
6. Educational systems use AI for personalized learning and online teaching support. AI helps students learn according to their individual needs and performance.
7. AI applications reduce manual work and increase automation in industries and offices. This improves productivity and saves time and operational costs.
8. AI supports research and data analysis by processing large amounts of information quickly. Scientists and businesses use AI for better predictions and planning.
9. Despite many advantages, AI also creates challenges such as privacy concerns, job displacement, and security risks. Proper monitoring and ethical use of AI are important.
10. Overall, AI has a strong positive impact on real-world domains by improving efficiency, accuracy, and innovation. It continues to transform industries and modern technology systems.